Rajat Sharma

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Portfolio and Profiles

Work Portfolio| GitHub Profile | LinkedIn Profile |Kaggle

Professional Summary

Working as a Data Scientist with 4+ years of experience delivering AI/ML and GenAI solutions that improve customer experience, operational efficiency, and revenue. Proven expertise in Python, SQL, Scikit-learn, PyTorch, RAGs, Agents, and building production-grade models. Achieved tangible results like multi billions cost saving, on ground reduction in escalations and better customer experience. Passionate about translating complex business problems into scalable data products.

Skills and Tools

- Languages: Python, SQL
- ML/DL Frameworks: Scikit-learn, PyTorch, Keras, Regression, Classification, Time Series
- GenAI: Langchain, Huggingface, RAGs, Agents & Agentic Workflows, LLM fine-tuning
- Data Viz: Tableau, Excel, PowerPoint
- Databases & Cloud: SQL (advanced), Google Cloud(GCP, Vertex Ai), Git/GitHub/GitLab
- Other: DataRobot, NLP, Deployment, Streamlit, Docker, Snowflake, ML-Ops

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Experience and Projects Discussion

Senior Analyst/Data Scientist | Dentsu Global Services | Sept 2024 – Present

Search and Rescue of Vulnerable orders (Classification, Gemini LLM, ML-Ops, Vertex AI Pipelines):

- Aim: Identify high-risk orders early by combining operational metrics and conversational signals to prevent executive escalations.
- Value: Significant reduction in executive escalations and cancellations, better customer experience
- Process: Have a starting base of faulty orders, aggregate it at an order level granularity with an anchor-date, implementing scaling and imputation, experiment and train various models with cross validation and holdout sets. Implement it using Vertex AI pipelines to get probability score from trained model and add conversational score (using Gemini for call and chat transcripts) on top of it to get the most risky orders.
- Results: Through A/B testing, achieved notable drop in escalations & cancellations while improving resolution time.
- Challenge: Preventing data leakage, handling severe class imbalance, and balancing LLM cost with accuracy.

Customer Clustering (Clustering, Statistical Analysis):

- Aim: Segment customers into cohorts based on behaviour to identify fraud and focus on profitable ones.
- Value: Eliminates fraud, enhances targeting of high-value customers.
- Process: Monitored purchasing habits, progressed to full-fledged modelling.
- Result: Model in progress, accuracy not finalized.
- Challenge: Detecting fraud customers creating new accounts post-ban remains unresolved.

Text Extraction from Images (GenAI, OCR,Streamlit)

- Aim: Extract SKU descriptions and quantities from unclear customer notes, map to system SKUs, and export to Excel.
- Value: Streamlines order processing by automating text extraction and SKU matching.
- Process: Pre-processed images, enhanced prompts with Gemini 2.5, converted results to Data Frame, and performed similarity matching with internal SKUs, returning top 2 matches.
- Result: Model achieved automated extraction with more than 95% accuracy.
- Challenge: Illegible notes with symbols, short hands, numeric, and dimensions complicated text recognition and matching.

Severity Calculation for Customer Pain Points (Huggingface, GenAI, Ranking, Finetuning)

- Aim: Tackle the most severe pain points to customer on priority hence give weightage accordingly.
- Value: Adds to better CX hence customer retention and better handling of issues.
- Process: Attaching pain points to the contacts made from Level_1(very broad) to Level_5(most granular) this was done using fine-tuned MPnet model for the pre decided pain point levels (given by business owner).
- Process: Creating metrics like cancellations and refunds, no of contacts and touches, span of contacts, customer feedback score, and escalation, sentiment to compare and calculate severity of a pain point. Then making their severity scores at L5 levels using equal weightage to all the features and then giving weightages based on feature importance, finally discussing with business.

<u>Associate Data Scientist | Factspan Analyst | June 2022 – Aug 2024</u>

Turn Around Time Prediction (Regression, Multi-Dimensional Data):

- Aim: Predict time between surgeries in operation rooms.
- Value: Boosts team efficiency in surgical scheduling.
- Process: Used extreme gradient boosting for regression on a complex dataset.
- Result: 70% accuracy, with 10–20-minute error margin.
- Challenge: Managing 300+ features, millions of records, and hierarchical variables.

Refill Request Volume Forecast (Time Series Forecasting, Feature Engineering,):

- Aim: Forecast daily pharmacy refill request volume.
- Value: Improves efficiency, inventory, and customer experience.
- Process: Applied SARIMA for time series forecasting with feature engineering.
- Result: 92% accuracy, MAE of 60 for ~600 daily requests.
- Challenge: Client wanted to limit error in 10% range.

Pyxis Min Max Levels Prediction (Time Series, Fbprophet):

- Aim: Predict min/max drug levels for Pyxis (medicine dispensing machines).
- Value: Optimizes inventory for high-cost medicines.
- Process: Used FB Prophet for top 50 medicines, after testing moving averages.
- Result: 85% mean accuracy with 90% confidence intervals.
- Challenge: Handling predictions for over 2,000 medicine series.

Radiology Care and Coordination (GenAI)

- Aim: Detect patient severity and incidental findings from radiology reports, provide alerts, and trend analysis.
- Value: Enhances patient care, automation saves costs.
- Process: Used Open AI endpoint with data masking for privacy, built a dashboard.
- Result: Delivered dashboard with severity trends and cost savings.
- Challenge: Ensuring data privacy and accurate keyword-based severity detection.

Subject Matter Expert (Chegg India) | Dec 2018 - Jun 2020 (Freelance Electrical & Electronics Eng) The key responsibilities revolved around helping students with their doubts and assignments (A freelance online work profile).

Education:

- **Masters** - MTech, NIT ALLAHABAD, MNNIT ALLAHABAD
- **Bachelors –** BTech, AKTU, ABES Engineering College

Extra Activities and Interests

- An active super-host on Airbnb with multiple listed properties.
- Stock market analysis and active investing/trading.